

■ 3.5.10 Manipulating Integrals in Symbolic Form

When *Mathematica* cannot give you an explicit result for an integral, it leaves the integral in a symbolic form. It is often useful to manipulate this symbolic form.

Mathematica cannot give an explicit result for this integral, so it leaves the integral in symbolic form.

```
In[1]:= Integrate[x^2 f[x], x]
Out[1]= Integrate[x^2 f[x], x]
```

Differentiating the symbolic form gives the integrand back again.

```
In[2]:= D[%, x]
Out[2]= x^2 f[x]
```

Here is a definite integral which cannot be done explicitly.

```
In[3]:= Integrate[f[x], {x, a[x], b[x]}]
Out[3]= Integrate[f[x], {x, a[x], b[x]}]
```

This gives the derivative of the definite integral.

```
In[4]:= D[%, x]
Out[4]= -(f[a[x]] a'[x]) + f[b[x]] b'[x]
```

Here is a definite integral with end points that do not explicitly depend on *x*.

```
In[5]:= defint = Integrate[f[x], {x, a, b}]
Out[5]= Integrate[f[x], {x, a, b}]
```

The partial derivative of this with respect to *x* is zero.

```
In[6]:= D[defint, x]
Out[6]= 0
```

There is a non-trivial total derivative, however.

```
In[7]:= Dt[defint, x]
Out[7]= -(Dt[a, x] f[a]) + Dt[b, x] f[b]
```